Shubhadeep Mandal

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Education

International Institute of Information Technology, Hyderabad — July 2025 – Present
Telangana, India
Master of Technology in Computer Science and Engineering — *CGPA: 9.12*

Indian Institute of Information Technology, Ranchi — Aug 2020 – Jun 2024
Ranchi, India
Bachelor of Technology (Hons.) in Computer Science and Engineering — *CGPA: 9.05*

Research & Open-Source Experience

Core Contributor, RookDB — Aug 2026 – Present
Independent Study, DSAC Lab, IIIT Hyderabad | Rust, Query Optimization, Testing — *Hyderabad, India*

- Co-architected the lab's open-source SQL-99 RDBMS in Rust across storage, parser and CLI crates, with **1,200+ tests**.
- Added a binder resolving every name and type before planning; differential testing matches SQLite on **50,000 random queries**.
- Built column statistics and a cost model choosing hash vs. index joins and join order, exposed via `EXPLAIN ANALYZE`.
- Wrote 100K–10M-row stress benchmarks whose fixes cut the 1M-row suite **14x (827 s to 58 s)**; scans hit **20M rows/s**.

Projects

Peer-to-Peer Distributed File Sharing System | *C++, Multithreading, Sockets, OpenSSL, Linux*

- Built a BitTorrent-style P2P system in C++; self-scheduling worker threads and a fix for a Nagle/delayed-ACK stall in length-prefixed TCP framing cut 1GB single-peer downloads from **193s to under 13s**.
- Removed a data race and a file-wide lock by confining shared state to one mutex each and writing out-of-order chunks with lock-free positional `pwrite`, one writer per piece, after a SHA-1 check of every 512KB piece.
- Architected Primary-Backup tracker failover with race-free election and state sync; **46 unit tests** pass under ASan/UBSan.

FOOL: Custom POSIX-Style Interactive Shell | *C++, POSIX APIs, Job Control, Signals, PTY Testing*

- Built a C++ shell with N-stage pipelines, ordered descriptor redirection (`2>&1`) and full job control (process groups, terminal handoff, Ctrl+Z, `fg`/`bg`), verified by a **138-check** suite including real-PTY keystroke tests.
- Eliminated signal races and lost exit statuses with the self-pipe trick for async-safe signals and reaping on the main thread.
- Cut startup **52x** (5.2s to 0.1s) by reading file types from directory entries instead of a `stat()` per file; built a zero-dependency raw-mode line editor with tab autocompletion and persistent history.

Warehouse Traffic Control: Distributed Robot Simulator | *JavaScript, Three.js, Web Workers*

- Built a deterministic 3D simulator of a 200-robot warehouse whose components talk only over a lossy, delayed, fault-injected network.
- Designed a lease-and-fencing protocol across 16 region managers, sustaining **0 collisions over 416K moves** under 40% message loss, ±20% clock drift and injected crashes.
- Implemented checkpoint and action-log replay enabling **sub-0.2 s time-travel seeks**, verified by headless state-equivalence tests; ran 56 benchmark simulations in parallel on Web Workers.

Technical Skills

Languages: C/C++, JavaScript, Rust, Python, SQL, Bash.

Software Engineering: OOP, Data Structures & Algorithms, Unit, Integration & Differential Testing, Benchmarking.

Systems: Multithreading & Synchronization, Processes & Signals, Sockets, Memory Management, Linux.

Tools: Git, Make, strace, ASan/UBSan, OpenSSL, Three.js, Docker.

Achievements

- **B.Tech Rank Holder:** Ranked 2nd in IIIT Ranchi by GPA in B.Tech CSE.
- **PGEE 2025:** Secured All-India Rank 100 in IIIT-Hyderabad Post Graduate Entrance Examination (PGEE).
- **Megathon 2025:** Third rank team in Megathon 2025 in the mobile computing track.

Interests

- Web Novel and Manga Enthusiast.
- Chess and Badminton.